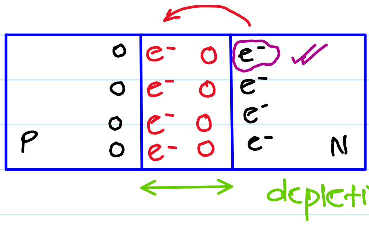
Electronics | Lecture # 02 | ECAT | MDCAT | Bilal Zia

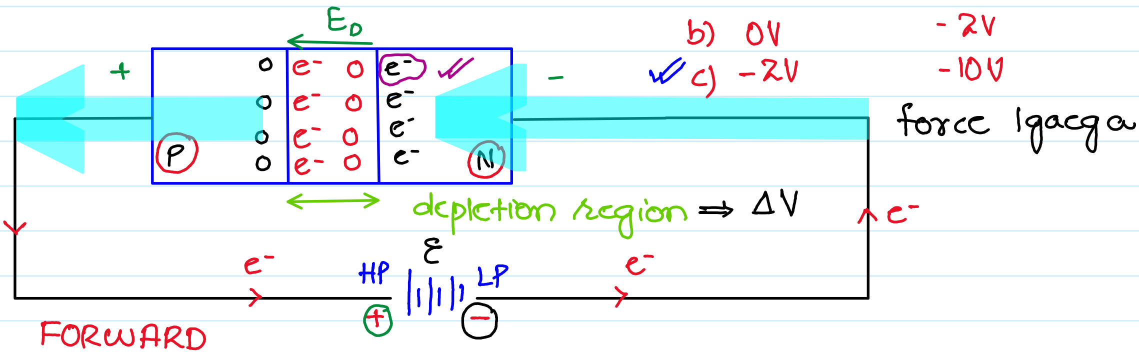
\# DIODE



\# BIASING

	HP	LP
a)	2V	0V
b)	0V	-2V
c)	-2V	-10V

Si → 0.7V
Ge → 0.3V
0.4V
- ε > 0.7V
- ε > 0.3V

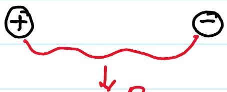


FORWARD

\# I = flow
\# Allow
\# depletion layer : THIN ≈ 0

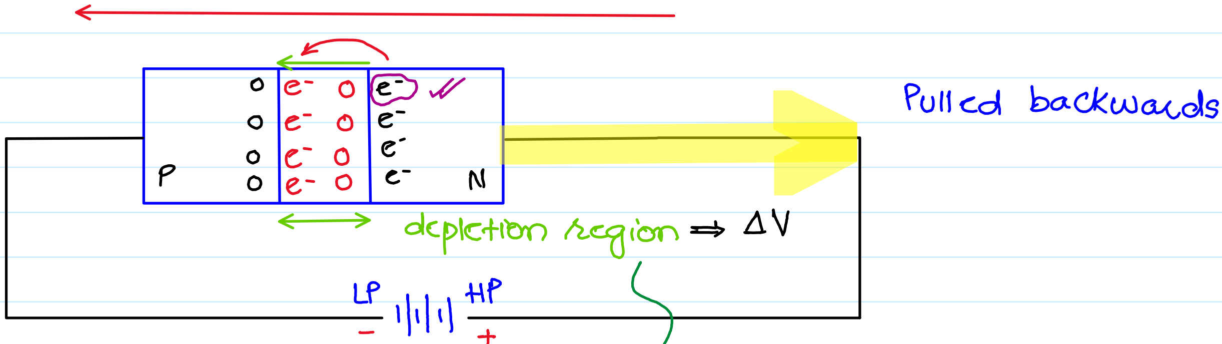
\# Short Ckt

R = 0



R = 0 ⇒ SHORT CKt path

\# REVERSE BIASING.



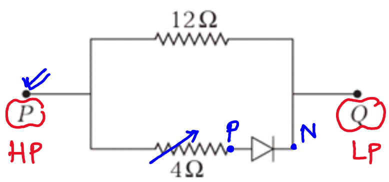
POTENTIAL BARRIER

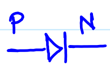
\# I = donot flow } BLOCK
\# diode : donot allow
\# Open CKt (R : ∞)

\# depletion layer : Width ↑↑

\# DIODE : ONE WAY SWITCH

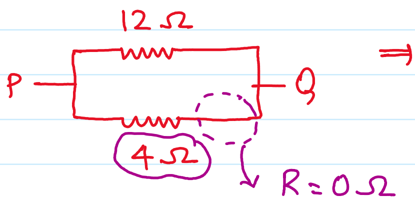
Find the net resistance of the network shown in the figure between the points P and Q, if (i) $V_P > V_Q$, (ii) $V_P < V_Q$.





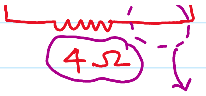
(i) $V_P > V_Q$

DIODE → fwd biased

⇒ 

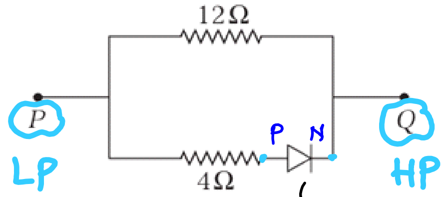
$\Rightarrow R_{eq} = \frac{R_1 R_2}{R_1 + R_2} \Rightarrow \frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2}$

$= 12 \times 4 / 16 = 3\Omega$

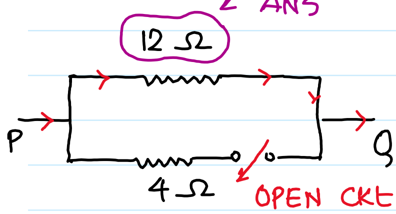


$$= \frac{12 \times \cancel{4}}{\cancel{16}\ 4} = 3\Omega$$

Find the net resistance of the network shown in the figure between the points P and Q, if (i) $V_P > V_Q$, (ii) $V_P < V_Q$.

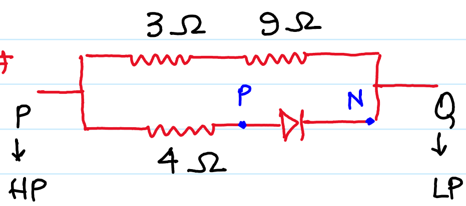


(2)



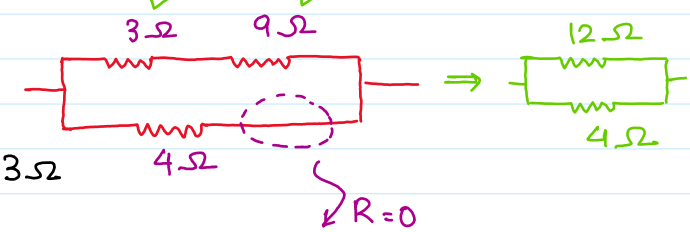
Rev. BIASED ⇒ diode ⇒ OC ⇒ R : ∞

Q#



(1) $V_P > V_Q$ (2) $V_P < V_Q$

TOTAL resistance

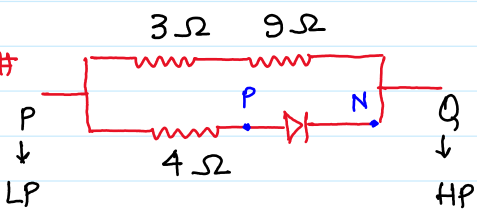


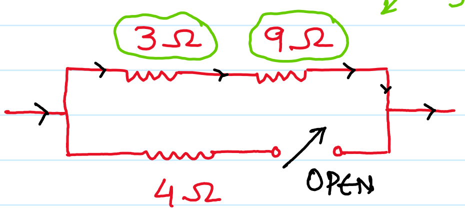
(1) $V_P > V_Q$ ⇒ diode : FWB ⟶ R : 0

$$\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2} \quad ; \quad R_{eq} = \frac{R_1 R_2}{R_1 + R_2} = 3\Omega$$

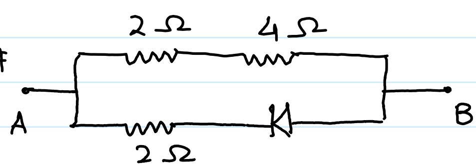
3Ω + 9Ω = 12Ω

Q#



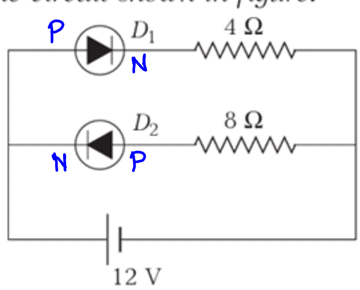


Q#



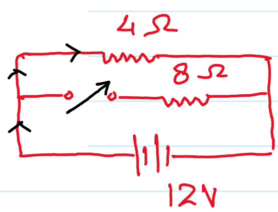
(1) $V_A > V_B$ (2) $V_A < V_B$

Find current passing through 4 Ω and 8 Ω resistance in the circuit shown in figure.



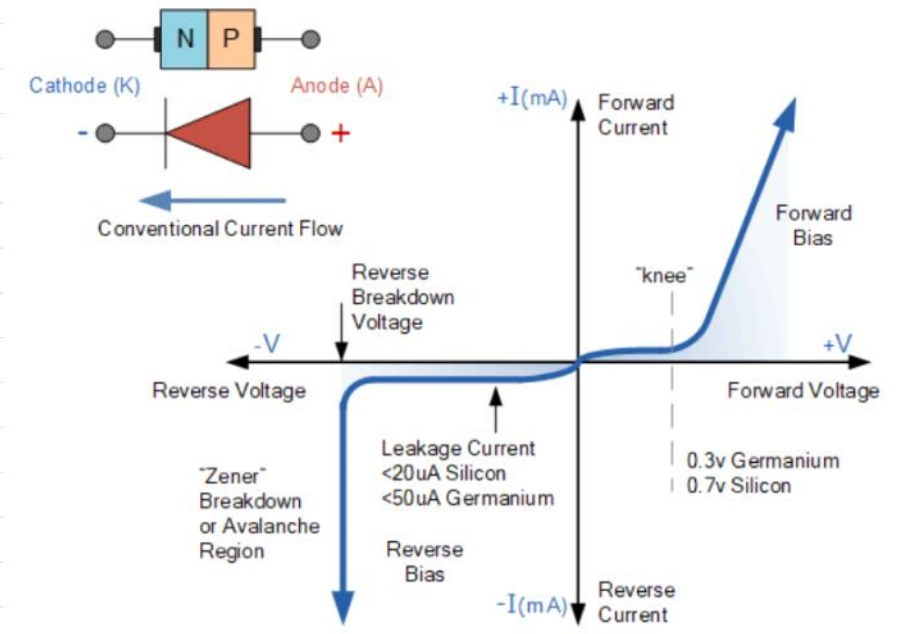
$D_1 \longrightarrow$ FWB ; $D_2 \longrightarrow$ RWB

8Ω ⟶ 0A



4Ω : $V = IR$

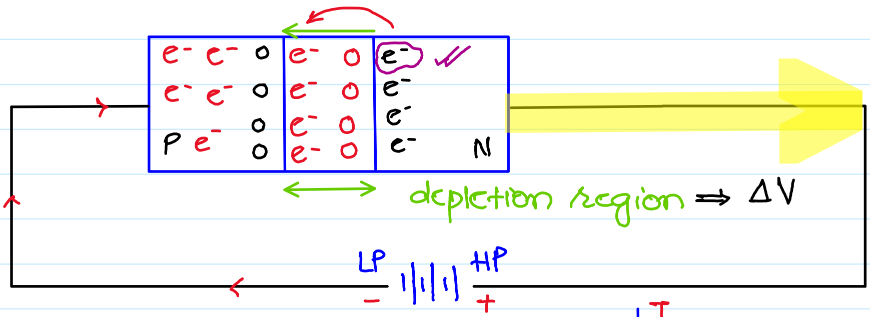
$$I = \frac{V}{R} = \frac{12}{4} = 3A$$



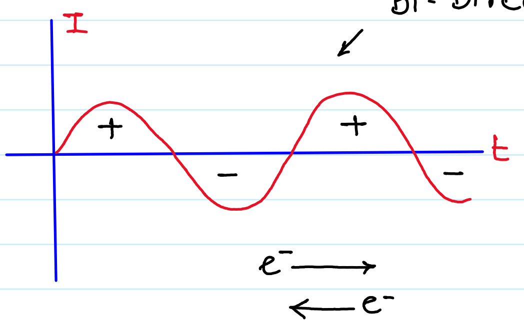
$V_A > P_B$

$V = IR \longrightarrow$ St line
$V \propto I \longrightarrow$ OHMIC Cond.

diode $\longrightarrow$ N. OHMIC
$\longrightarrow$ Non. linear



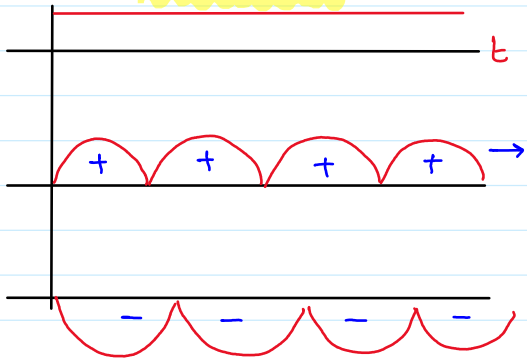
Bi-Directional I



RECTIFICATION

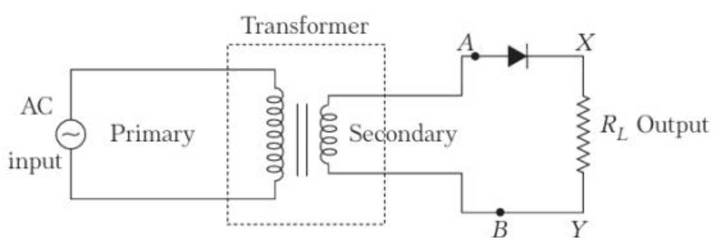
AC → DC

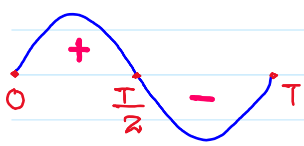
I Smooth DC



→ Ripples Wala
(Pulsating DC) → UNI-direction
↓
Smooth DC

A simple rectifier circuit called the half-wave rectifier, using only one diode is shown in figure. The AC voltage to be rectified is connected to the primary coil of a step-down transformer. Secondary coil is connected to the diode through resistor R_L across which, output is obtained.





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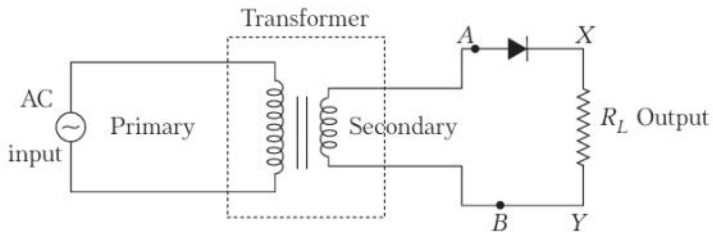
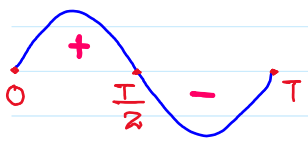


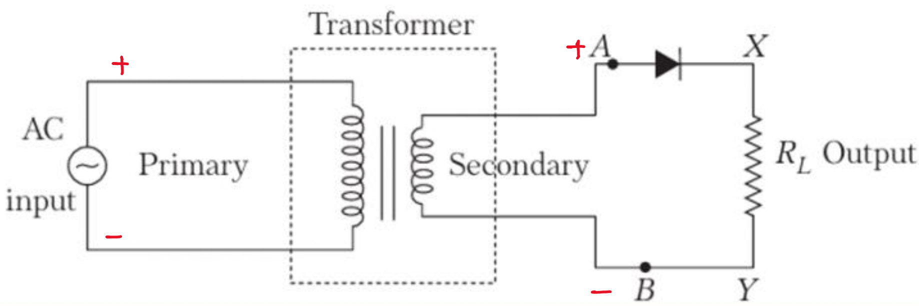


$0 \rightarrow T/2$: FIRST Half CYCLE

$T/2 \rightarrow T$: SECOND HALF CYC.

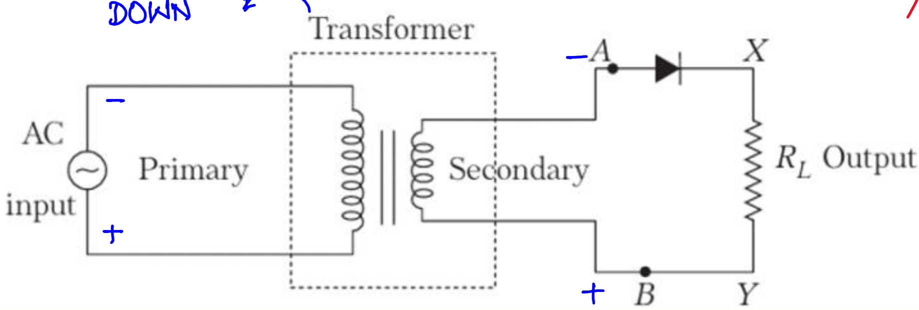
$0 \rightarrow T/2 \Rightarrow$ FWB

$\Rightarrow$ I · flow



$x \longrightarrow 3x$

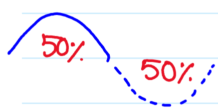
DOWN



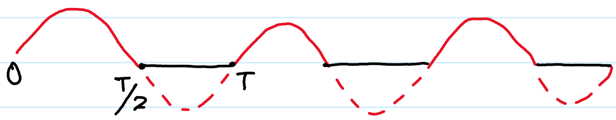
$T/2 \rightarrow T \Rightarrow$ RWB

$\Rightarrow I = 0$

allow



block



KCL

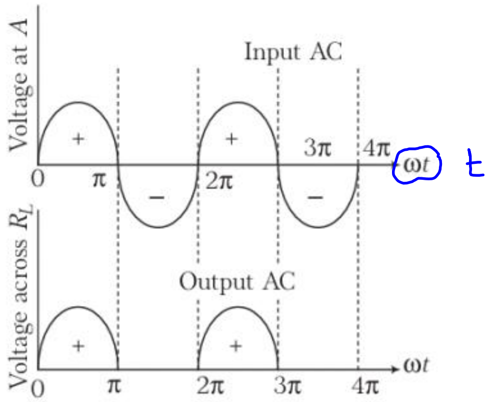
4A → 2A

E = VIt

50% 50%

Working

During positive half cycle of the input AC (*i.e.* $V_A > 0$), the p-n junction is in forward biased. Thus, the resistance in p-n junction becomes low and current flows. Hence, we get output in the load.

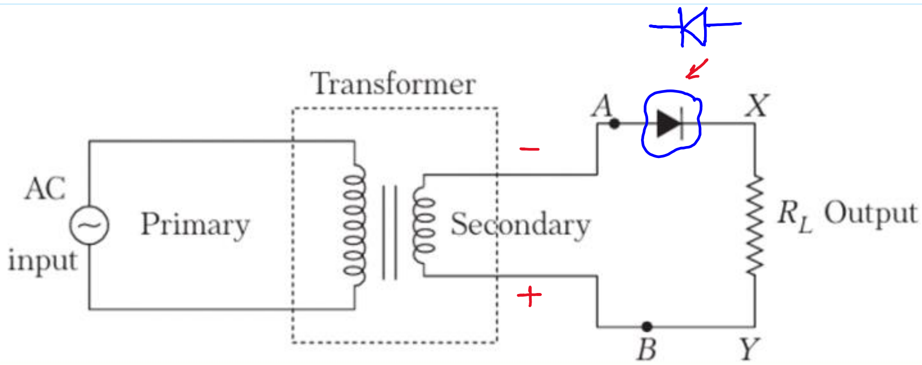


NATIONAL MDCAT → 2021

Ripple frequency → 50 rps

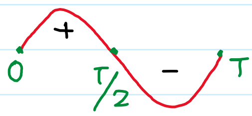
Wohe hogy jo source ke "freq"

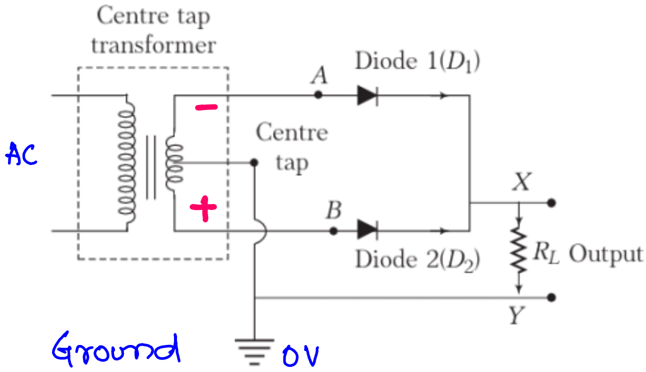
AC → 50Hz → 50 cyc/sec



(i) Centre tap full wave rectifier

Figure shows a circuit which is used in full wave rectification. Two diodes are used for this purpose.





0 – T/2 → FIRST HALF CYCLE

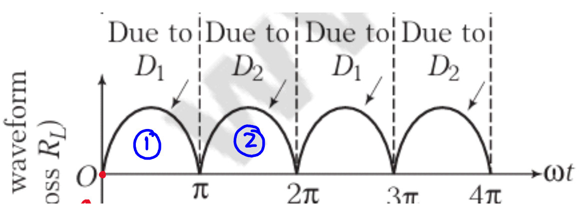
+ HP → D_1 → 0 LP ✓ FB — 5V (HP) — 0V (LP)

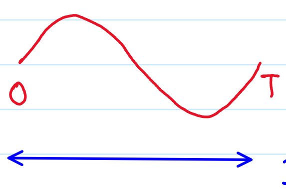
– LP → P N D_2 → 0 HP RB — -5V LP — 0V HP

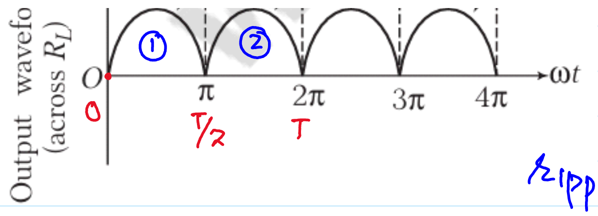
T/2 – T →

– LP → 0 HP — D1 = RB

+ HP → 0 LP — D2 → FB

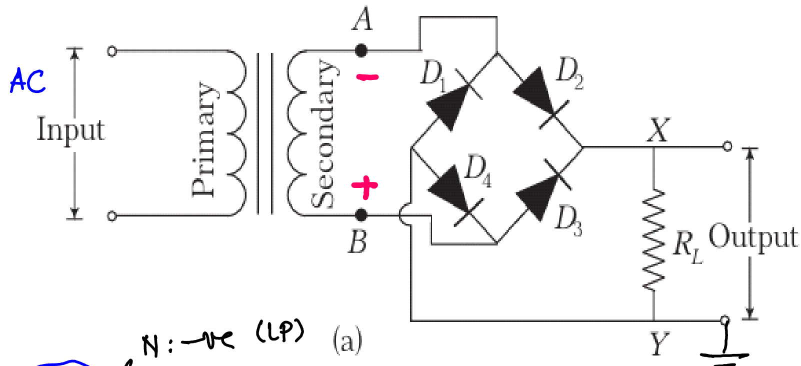






1 CYCLE

Ripple freq = 2 x source freq



(a)

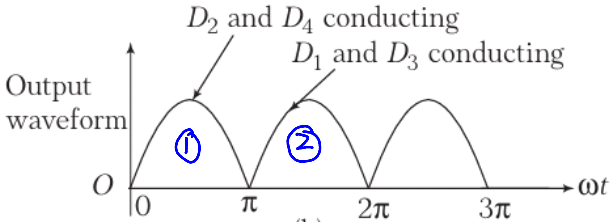
0 → T/2

D1: N : + HP; P : 0 LP — RB

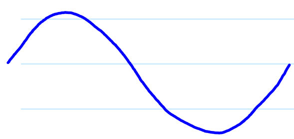
D3: N : 0 (HP); P : −ve (LP) : RB

D4: N : −ve (LP); P : 0 (HP) : FB

D2: N : 0 (LP); P : + (HP) : FB



(b)



In a full wave rectifier circuit operating from 100 Hz mains frequency, what is the fundamental frequency in the ripple?

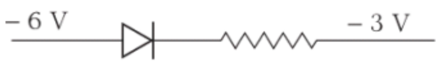
ripple

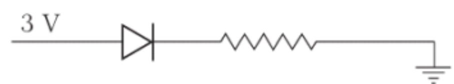
⇒ 2 x 100

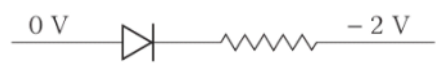
⇒ 200 Hz

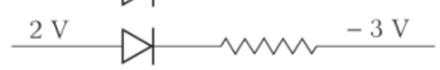
100 Hz

A reverse-biased diode is

(a) 

(b) 

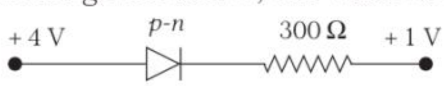
(c) 

(d) 

Number of electrons in the valence shell of a semiconductor is

(a) 1 (b) 2 (c) 3 (d) 4

In the circuit given below, the value of the current is



(a) zero (b) 10^{-2} A (c) 10^{2} A (d) 10^{-3} A

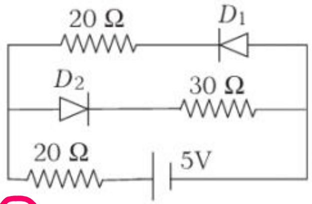
HINT $I = \frac{V}{R} = \frac{\text{Pot DIFF}}{R}$

$= \frac{(+4) - (+1)}{300}$

(a) zero (b) 10^{-2} A (c) 10^{2} A (d) 10^{-3} A

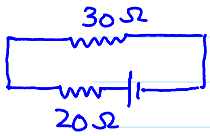
$= \frac{(+4)-(+1)}{300}$

The current in the circuit will be



(a) (5/40) A (b) (5/50) A (c) (5/10) A (d) (5/20) A

HINT D1 RB and D2 : FB

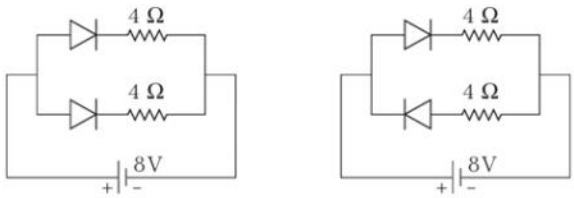


SERIES ckt

$\Rightarrow I = \frac{V}{R}$

$R_{eq} = 20+30$
$= 50\Omega$

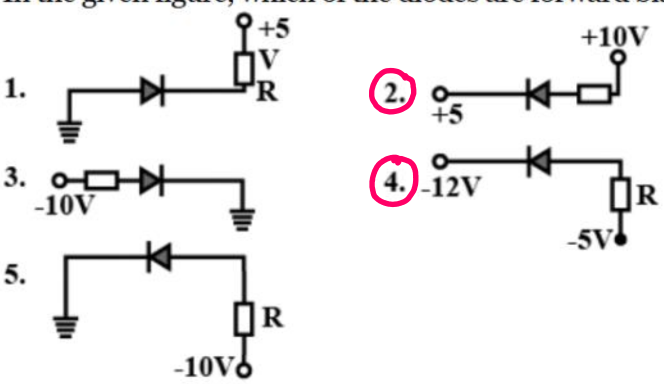
Currents flowing in each of the circuits A and B respectively are



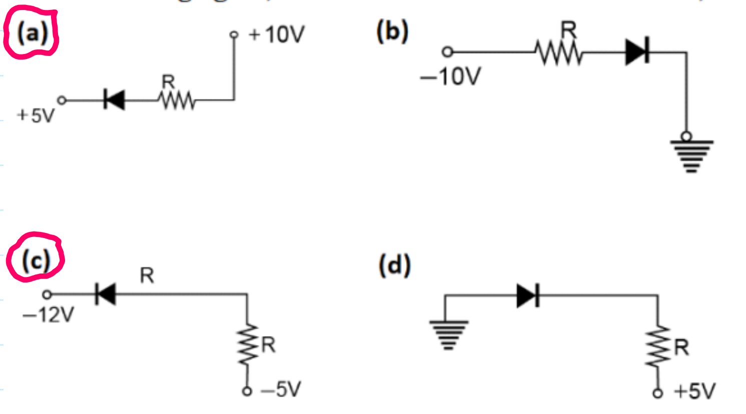
(Circuit A) (Circuit B)

(a) 1 A, 2 A (b) 2 A, 1 A
(c) 4 A, 2 A (d) 2 A, 4 A

In the given figure, which of the diodes are forward biased?



In the following figure, the diodes which are forward biased, are



A. (a), (b) and (d)
B. (c) and (a)
C. (b) and (d)
D. (c) only

Out of the following, which one is a forward biased diode?